23MIC0055 KISHORI KAMALI M

In [1]:
```
import pandas as pd
```

In [4]:
```
data=pd.DataFrame({'Country':['USA','NLD','SGP','NOR','AUS','CAN','CAN','USA','MEX','ZAf','ZAF','USA','DEU','CHE','
                   'City':['tampa','breda','singapore','oslo','sydney','vanouver','montreal','cambridge','xalapa','
                   'GVI':[36.1,29.3,29.3,28.8,25.9,25.9,25.5,25.3,24.0,23.7,23.6,23.6,21.5,21.4,21.4,20.6,20.0,19.5
                   'Rank':[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30]})
print(data)
```

	Country	City	GVI	Rank
0	USA	tampa	36.1	1
1	NLD	breda	29.3	2
2	SGP	singapore	29.3	3
3	NOR	oslo	28.8	4
4	AUS	sydney	25.9	5
5	CAN	vanouver	25.9	6
6	CAN	montreal	25.5	7
7	USA	cambridge	25.3	8
8	MEX	xalapa	24.0	9
9	ZAf	durban	23.7	10
10	ZAF	johanesburg	23.6	11
11	USA	sacramento	23.6	12
12	DEU	frankfurt	21.5	13
13	CHE	geneva	21.4	14
14	MEX	veracruz	21.4	15
15	NLD	amsterdam	20.6	16
16	USA	seattle	20.0	17
17	CAN	toronto	19.5	18
18	USA	miami	19.4	19
19	USA	boston	18.2	20
20	ISR	telaviv	17.5	21
21	MEX	guadalarajara	17.3	22
22	MEX	leon	16.5	23
23	ITA	turin	16.2	24
24	USA	losangeles	15.2	25
25	ARG	buenos_aires	14.5	26
26	BRA	franca	13.7	27
27	USA	new_york	13.5	28
28	ZAF	capetow	13.4	29
29	GBR	london	12.7	30

AVERAGE GVI OF USA

In [11]:
```
avg_data=(pd.pivot_table(data,values="GVI",index="Country",aggfunc="mean"))
print(avg_data.loc['USA'])
```

```
GVI    21.4125
Name: USA, dtype: float64
```

MINIMUM OF GVI OF NETHERSLANDS

In [13]:
```
min_data=pd.pivot_table(data,values="GVI",index="Country",aggfunc="min")
print(min_data.loc['NLD'])
```

```
GVI    20.6
Name: NLD, dtype: float64
```

SUM OF GVI[CITIES OF ITALY]

In [26]:
```
sum_data=pd.pivot_table(data,values="GVI",index="Country",aggfunc="sum")
print(min_data.loc['ITA'])
```

```
GVI    16.2
Name: ITA, dtype: float64
```

FIND THE CITY AND COUNTRY OF LEAST GVI

In [21]:
```
mi_data=pd.pivot_table(data,values="GVI",index=["Country","City"],aggfunc="min")
index=data['GVI'].idxmin()
print(data.loc[index,"City"],data.loc[index,"Country"])
```

```
london GBR
```

FIND THE COUNTRY THAT HAS MAXIMUM NO OF CITIES

In [22]:
```
count_data=(pd.pivot_table(data,values="GVI",index="Country",aggfunc="count"))
print(count_data)
```

```
         GVI
Country
ARG        1
AUS        1
BRA        1
CAN        3
CHE        1
DEU        1
GBR        1
ISR        1
ITA        1
MEX        4
NLD        2
NOR        1
SGP        1
USA        8
ZAF        2
ZAf        1
```

In [30]:
```
index=count_data['GVI'].idxmax()
print(f"the country that has maximum no of cites is {index} ")
```

```
the country that has maximum no of cites is USA
```

In [ ]:
```

```